A (Mostly Qualitative) Study of Modularity Robustness to Species Introduction in Plant-Pollinator Networks

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Introduction

- Plant pollinator networks
  - Bipartite graph: plants and pollinators
  - Interactions: observed in field, pollen counts
- Modularity
  - Nodes clustered into groups (modules)
  - High connectivity within groups, low connectivity outside
  - Appears in plant-pollinator networks (Olesen et al., 2007)
  - Related to persistence, robustness, coevolution (Stouffer, Bascompte, 2009)

Modularity score of a partition into k modules:

$$M = \sum_{s=1}^{k} \left(\frac{l_s}{L} - \left(\frac{d_s}{2L} \right)^2 \right)$$

Robustness to Invasion

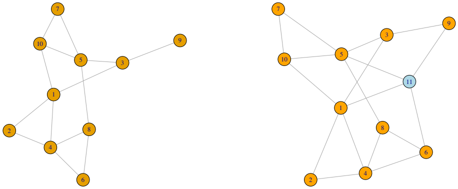
- Invaders reshape communities all the time
- Invasive plants could be good or bad for natives (Travaset, Richardson, 2014)
- Invasive plants often supergeneralists, becoming network hubs (Aizen et al., 2008) (Travaset et al., 2013)
- Modularity usually lower in invaded networks

Studies of modularity thus far have studied networks *after* invasion.

How do plant invasions affect the modularity of a network?

Experiment

- Introduce one node (plant) to existing data set
- Study change in modularity under various conditions
- Study difference in module assignment



Measuring Distances Between Modules

$D(C, C')$, variation (Karrer et al., 2008) on metric devised by van Dongen (2000). For two separate group assignments (clusterings) C, C', where there are K clusters in C and K' clusters in C',

$$D(C, C') = 1 - \frac{1}{2n} \left(\sum_{k=1}^{K} \max_{k'} \left(|C_k \cup C'_{k'}| \right) + \sum_{k'=1}^{K'} \max_{k} \left(|C_k \cup C_{k'}| \right) \right)$$

The two summands are row- and column-maxima of a $k \times k'$ confusion matrix.

Example of Distance Calculation

Consider the following clusterings:

	A	B	C	D
C	$\{1,3,8,10\}$	$\{4,6\}$	$\{5,9\}$	$\{2,7\}$
	A	B	C	
C'	$\{1,2,3,9,10\}$	$\{4,5,6,8\}$	$\{7\}$	

The confusion matrix is

$$N = \begin{array}{c c} & \begin{array}{ccc} A & B & C \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \end{array} & \begin{pmatrix} 3 & 1 & 0 \\ 0 & 2 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \end{array}$$

$$D(C, C') = 1 - \frac{1}{20}(3+2+1+1+3+2+1) = 0.35$$

How to Integrate Invasive Species?

Use five different incorporation schemes (i.e., edge assignments)

1. Randomly. Select number of new edges m, randomly assign to all pollinators in network.
2. Integrate within a module. Assign edges to pollinators within module with $P = p_{in}$, assign edges to pollinators in other modules with $P = p_{out}$, where $p_{in} > p_{out}$.
3. Assign edges with probability proportional to number of observed interactions of a pollinator (i.e., sum of edge weights)
4. Assign edges with probability proportional to degree. (Traveset et al., 2013) (Stouffer et al., 2014)
5. Assign edges with probability inversely proportional to degree. (Aizen et al., 2008) (Stouffer et al., 2014)

Initial Methods

Software used:

- `R` (3.2.3)
- package `igraph`
- package `bipartite`

Data set used:

- Kato et al. (1990)
- 94 plants, 679 pollinators
- Modularity score: 0.611, 3.8% connected

Procedure:

- For each incorporation scheme, run 500 trials
- Add one node, assign links
- Record number of links added, ΔM, D

Change in Modularity Score

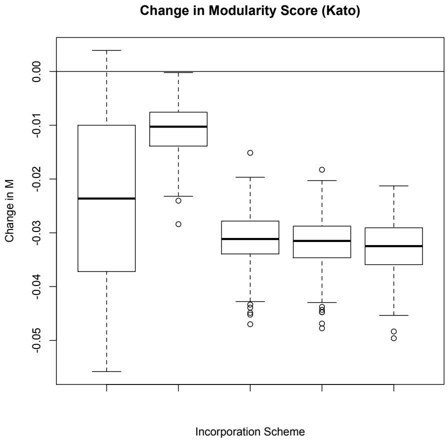


Figure 1: Boxplots of change in modularity score with different incorporation schemes. In each case, $n = 500$

Change in Modularity Score

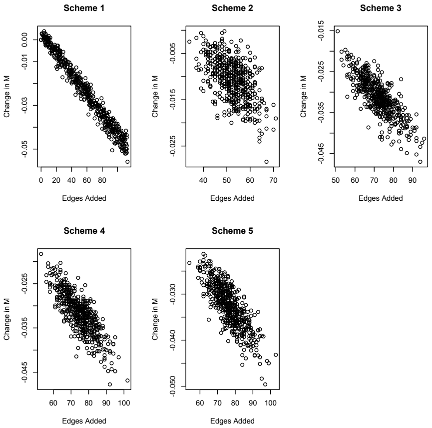


Figure 2: Plot of change in modularity score against number of edges added (e.g., generality of plant).

Change in Modularity Score

Scheme	b_1	p	R^2
1	$-4.65 * 10^{-4}$	0	0.971
2	$-3.81 * 10^{-4}$	0	0.300
3	$-4.76 * 10^{-4}$	0	0.648
4	$-4.67 * 10^{-4}$	0	0.655
5	$-5.11 * 10^{-4}$	0	0.689

Table 1: Summaries of regressions of change in modularity score against edges added in Kato dataset.

Distances Between Partitions

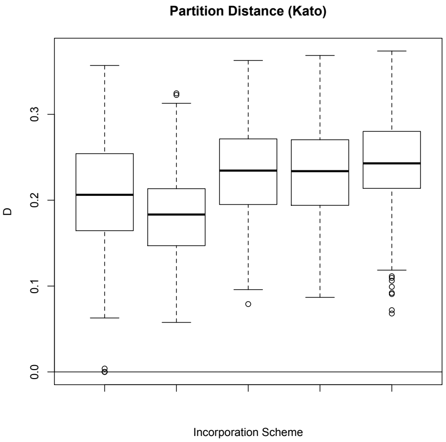


Figure 3: Boxplots of partition distance with different incorporation schemes. In each case, $n = 500$.

Distances Between Partitions

Scheme	b_1	p	R^2
1	$+9.93 * 10^{-4}$	0.000	0.269
2	$-2.21 * 10^{-5}$	0.946	0.000
3	$+8.44 * 10^{-4}$	0.003	0.015
4	$+6.50 * 10^{-4}$	0.028	0.009
5	$+2.81 * 10^{-4}$	0.352	0.001

Table 2: Summaries of regressions of partition distance against edges added in Kato dataset.

Redistributed Interactions

- Connectivity of invaded networks no different than non-invaded networks (Travaset et al., 2013) (Stouffer et al., 2014)
- Network-level changes due to redistribution of interactions (Aizen et al., 2008)

Add new step: randomly delete links between new partner of invasive plant with native plants (with varying probability: $p_r = 0.10, 0.25, 0.50$)

Changes in Modularity Score with Edge Deletion

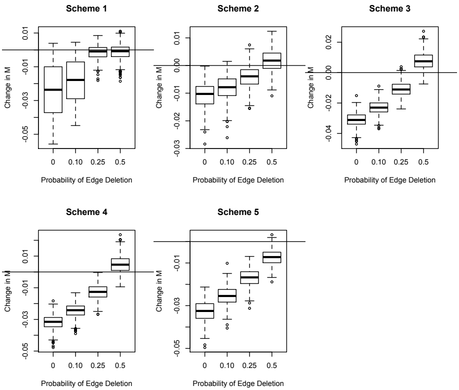


Figure 4: Changes in modularity score with varying probability of deleting edges between invasive plant partners and preexisting plants.

Changes in Modularity, Effects of Invader Degree

	$p = 0$		$p = 0.10$		$p = 0.25$		$p = 0.50$	
Scheme	b_1	R^2	b_1	R^2	b_1	R^2	b_1	R^2
1	$-4.6e^{-}4$	0.97	$-3.7e^{-}4$	0.94	$-2.4e^{-}4$	0.82	$-6.4e^{-}5$	0.22
2	$-3.8e^{-}4$	0.30	$-3.0e^{-}4$	0.19	$-2.1e^{-}4$	0.10	$-3.2e^{-}5$	0.00
3	$-4.7e^{-}4$	0.65	$-4.4e^{-}4$	0.51	$-3.9e^{-}4$	0.31	$-2.1e^{-}4$	0.05
4	$-4.7e^{-}4$	0.66	$-3.9e^{-}4$	0.43	$-3.4e^{-}4$	0.25	$-2.0e^{-}4$	0.05
5	$-5.1e^{-}4$	0.69	$-4.5e^{-}4$	0.60	$-3.3e^{-}4$	0.35	$-1.3e^{-}4$	0.05

Table 3: Summaries of regressions of modularity change against edges added in Kato dataset, with various probabilities of edge deletion.

Changes in Modularity Distance with Edge Deletion

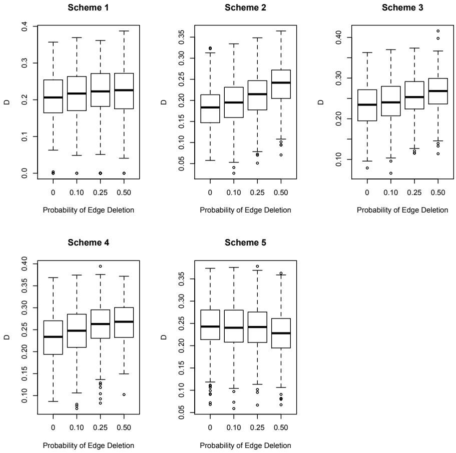


Figure 5: Distance of the new optimal partition from original with varying probability of edge deletion.

Do the observed patterns above hold in other networks?
For comparison:

1. Junker et al., 2013
   - 56 plants, 257 pollinators
   - Modularity 0.515, 7.9% connected
2. Kevan, 1970
   - 30 plants, 114 pollinators
   - Modularity 0.413, 18.2% connected

Change in Modularity in Different Networks

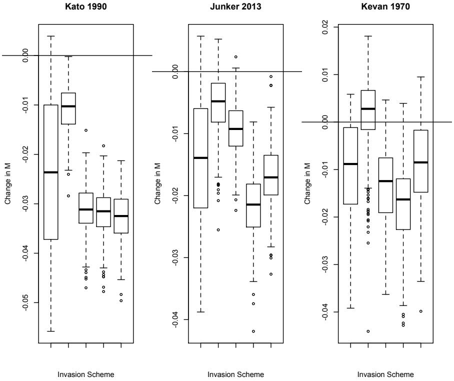


Figure 6: Change in Modularity Score of different incorporation schemes with edge addition.

Partition Distances in Different Networks

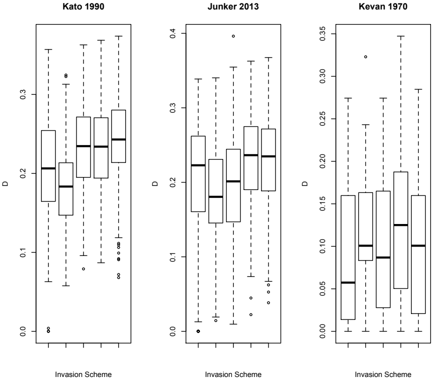


Figure 7: Change in Modularity Score of different incorporation schemes with edge addition.

Modularity in Different Networks, With Edge Removal

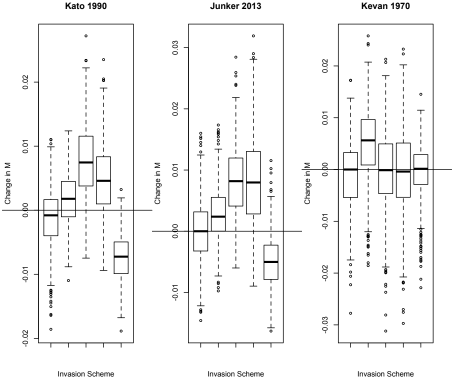
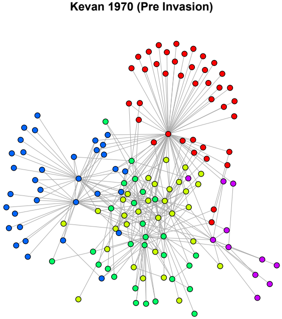
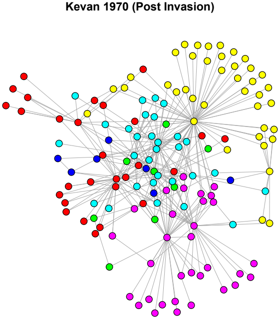


Figure 8: Change in modularity score in different networks with different incorporation schemes. These trials all included edge removal with 50% probability.

An Example





A Qualitative Look Inside

	k..mc2															
k..mc	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	0	0	0	1	86	0	0	0	0	0	0	0	0	0	0	0
2	0	0	42	0	0	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0	40	0	1	0	0	0	0	23	0	0	0	0	0
4	45	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	1	0	0	0	46	0	12	0	0	0	0	1	0	0	0
6	1	0	0	0	0	0	0	0	29	0	0	0	9	0	0	0
7	0	33	0	1	0	0	0	0	0	0	3	0	0	0	0	0
8	0	1	0	0	0	0	0	1	0	0	24	2	0	0	0	0
9	0	0	0	0	0	0	82	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	1	0	55	0	0	1	0	0	0	0	0
11	0	0	0	0	3	20	0	0	0	0	0	14	0	0	0	0
12	0	0	0	0	0	0	0	0	0	131	0	0	0	0	0	0
13	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14	0	0	0	0	0	0	0	0	0	0	0	0	39	0	0	0
15	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0
16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
17	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Conclusions (Math)

- Degree of added node has an effect on modularity when edges are added
- When edges of the added node's neighbor are deleted, this relationship becomes weaker
- Modularity may increase in cases with high edge removal
- The way the invasive node is wired into the population affects modularity
- In modular networks, even small introductions can cause large shifts in module composition (large variation)
- Less modular networks are have more robust partitions?

Implications (Biology)

- Increased modularity in mutualistic networks associated with lowered stability (Thebault, Fontaine, 2010)
- Decreased modularity of invaded networks means danger in removing invasives (Valdovinos et al., 2009)
- Modules could be associated with coevolutionary processes

With this in mind:

- An invasive generalist can increase overall network stability
- Stability effects less predictable as more pollinators abandon native hosts
- Large transfers of species between modules is possible, perhaps meaningless
- Other metrics (nestedness, asymmetry) also useful in assessing network health